

Multimedia Databases Exercises

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Exercise 7

Subject: Content-based retrieval

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Task 1: Content-Based Retrieval (CBR)

1. What does Content-Based Retrieval mean?
2. What are the components of a CBIR architecture? Explain the basic principle of each component using an example.
3. What is a feature vector?
4. Which problems occur when indexing feature vectors?

Task 2: CBR Terminology

Explain the following notions in your own words. Give an example each:

1. Dominant Color
2. Spatial Coherency
3. Distance Metrics
4. Curse of Dimensionality
5. Types of content-based queries

Task 3: CBR systems

Which necessary conditions must be fulfilled in order to issue the following query to a CBIR system: *"Give me all images which contain a blue truck"*. What are the problems that can occur?

Task 4: Image indexing by colours

The starting points are the following two images shown in figure 1. Supposing every image has a resolution of 4x4 pixels. Thus the left image contains exactly 8 blue coloured and 8 white coloured pixels, the right one one big black and one big white block:

1. Apply an even colour quantification for 8 colours. Which quantification area (range) do the colours in the two images belong to?
2. Create a colour histogram for both images.
3. Apply an even bin quantification for 2 bit.

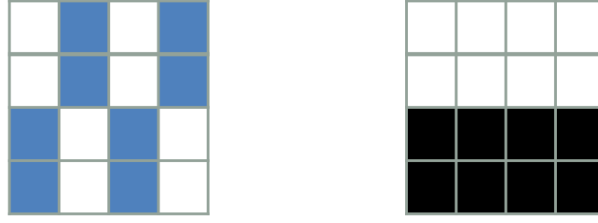


Figure 1: example figures

Task 5: Similarity of images because of colour distribution

1. **Minkowski Distances:** Use the two images of exercise 4 and the results of exercise 4.2. Determine the similarity of the images with the help of Minkowski distances L_1 , L_2 and L_∞ . Which result would you obtain if the red colour in the left image was black? Which conclusions do you draw?
2. **Non-parametrical Distances:** Starting with the following colour distribution: $H1 = (5, 5, 5, 5)$ and $H2 = (8, 5, 4, 3)$ Calculate the distances between $H1$ and $H2$ with the help of the following functions:
 - a) Kolgomorov- Smirnov Distance
 - b) Chi-squared Distance

Definitions

Minkowski Distance

Given $P = (x_1, x_2, \dots, x_n)$ and $Q = (y_1, y_2, \dots, y_n) \in R^n$

$$L_p(P, Q) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

Kolgomorov- Smirnov Distance

$$KS(P, Q) = \max_i |F^r(i; P) - F^r(i; Q)|$$

where $F^r(i; P)$ is equivalent to the kumulative histogram of P in place i .

Chi-squared Distance

$$D_\chi(P, Q) = \sum_i \frac{(x_i - f'(i))^2}{f'(i)}$$

where

$$f'(i) = \frac{x_i + y_i}{2}$$